MODEL CHANGE

Price now enters before bookings are drawn

NEW



$$\nu_k = \lambda \cdot \zeta_k \cdot \Sigma_j \, s_j(k) \cdot f_j(p_k / p_k^{std})$$

expected requests at lead k = demand level × pace shock × each segment's share of the booking curve × its response to the price ratio

Posted price p

The client's ADR ladder on expected fullness, times a random per-night level (sd 10%). Fixed before bookings arrive (open loop).

Reference price p^{std}

The standard price: the same ladder without the random level. The ratio p / p_std is the night's deviation from usual pricing, so v3's calibration is kept.

Segment response f_j

Each segment scales its requests by f(ratio). Elasticity at the standard price: group 1.8, leisure 1.2, last-minute 0.5 (assumed).

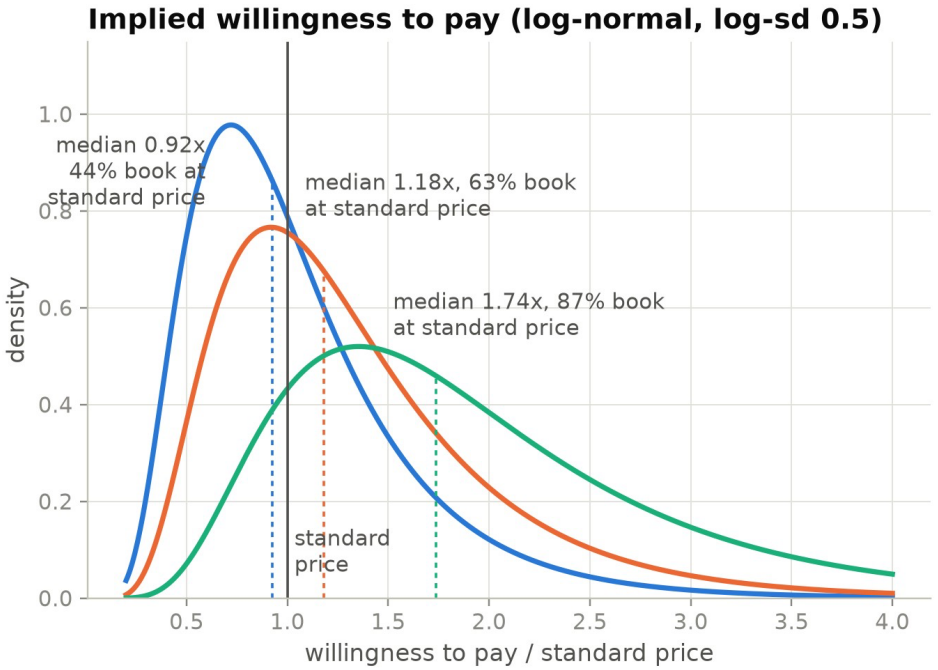
With price response switched off, the simulator output is identical to v3.

MODEL CHOICE

Two response models, same starting point

Both are calibrated to the same demand and the same elasticity at the standard price. They behave differently as price moves away from it.





Constant elasticity keeps sensitivity fixed at every price. A segment with elasticity below 1 (last-minute) earns more the higher the price, so there is no best price.

Willingness to pay becomes more sensitive as price rises. Implied median WTP: group 0.92×, leisure 1.18×, last-minute 1.74× the standard price. Separately, the ladder posts lower dollar prices early.

Assumed, not calibrated: elasticities 1.8 / 1.2 / 0.5 and WTP spread (log-sd 0.5).

EXPERIMENTAL RESULTS

How does price affect demand and revenue?



At 1.2× the standard price

E1: same hotel-nights, only price changed (10 hotels, 2023-2025)

	Constant elasticity	Willingness to pay
Expected demand	−18%	**−20%**
Rooms sold	−12%	**−14%**
Revenue	+7%	**+4%**

Both models cut demand when price rises. They differ on revenue: willingness to pay has a best price; constant elasticity is still rising at 2×.

E2: 10 hotels, 2024. Results depend on assumed elasticities (appendix A2). The 1.2× peak describes the simulator, not a pricing recommendation for real hotels.

What is verified and what is next

Implementation validation

Identical

Price response off: every output table, config and generator hash match v3 byte for byte

18 / 18

Your original checks pass in legacy and in price mode (one check is reinterpreted, appendix A1)

28 / 28

New tests pass: segment split sums to v3's curve, factor = 1 at the standard price, exact segment elasticities, booking and revenue accounting

Limitations and next steps

1 Reference price

Demand responds to deviations from the standard pricing policy, not to the standard price path itself. Next: separate the baseline reference policy from the actual posted prices, so any price path affects demand.

2 Closed-loop pricing

Prices currently follow expected occupancy (open loop). Next: let prices adapt to actual bookings as they arrive.

3 Price-aware cancellations

Cancellations are still modelled independently of price.

Elasticities (1.8 / 1.2 / 0.5) and the willingness-to-pay spread are assumed, not calibrated.